

Physical Chemistry Practical 4: Reaction Kinetics

Gas Syringe Volume Monitoring & Sodium Thiosulfate Disappearing Cross Rate Investigations

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Physical Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Continuous Monitoring of Reaction Rate: Magnesium and Hydrochloric Acid

Cambridge Syllabus Reference: 9701/35/M/J/23/Q2 · [20 Marks]

Chemicals and Apparatus Provided:

- FA 1 is 1.00 mol dm^{-3} hydrochloric acid, HCl.
- Magnesium ribbon, Mg (cut into clean 5.0 cm strips, cleaned with emery paper).
- Conical flask (100 cm^3) fitted with a bung, delivery tube, and 100 cm^3 gas syringe.
- Stopwatch.

Experimental Procedure:

1. Using a measuring cylinder, pour 25.0 cm^3 of 1.00 mol dm^{-3} HCl (FA 1) into the conical flask.
2. Set up the gas syringe horizontally with the plunger set to 0.0 cm^3 .
3. Add one 5.0 cm strip of magnesium ribbon to the flask, immediately insert the rubber bung, and start the stopwatch.
4. Record the volume of hydrogen gas collected in the gas syringe every 10 seconds for 2 minutes.

(a) Record your gas volume readings in the table below at 10-second intervals from $t = 0$ to $t = 120 \text{ s}$. [4]

Time / s	0	10	20	30	40	50	60	75	90	105	120
Vol / cm^3	0.0										

(b) On the grid provided, plot a graph of volume of hydrogen gas (vertical axis) against time (horizontal axis). Draw a smooth curve of best fit through the points. Draw a tangent to the curve at $t = 0 \text{ s}$ and calculate the initial rate of reaction. [6]

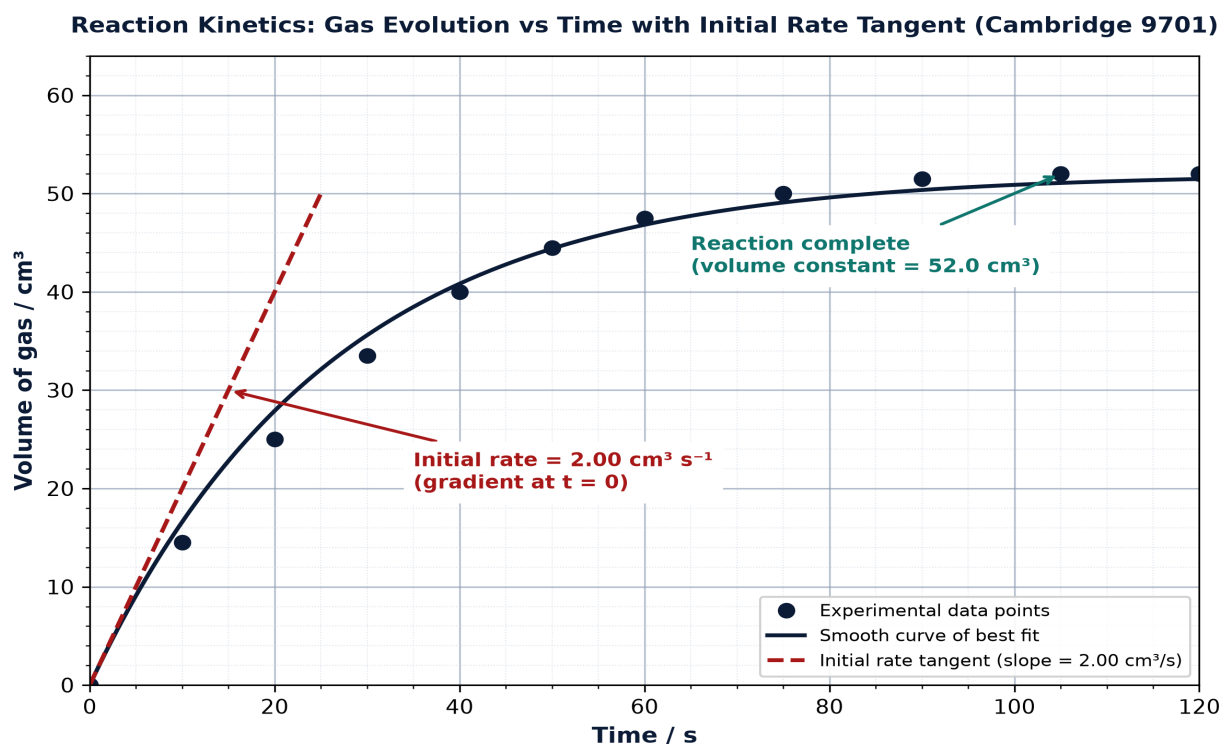


Fig 1.2: Gas evolution curve with tangent at origin to calculate initial rate.

(c) State the value of the initial rate calculated from your tangent. [1]

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(d) Why does the rate of reaction decrease continuously as time progresses? [2]

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(e) State the maximum volume of hydrogen gas collected, and calculate the amount, in moles, of H₂ produced.
[Molar gas volume at r.t.p. = 24.0 dm³ mol⁻¹] [2]

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(f) $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$. Calculate the mass of magnesium that reacted. [Ar: Mg = 24.3] [2]

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(g) State one source of systematic error when inserting the bung and how the apparatus could be modified to eliminate this error. [3]

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Question 2: Disappearing Cross Kinetics: Effect of Concentration on Sodium Thiosulfate Rate

Cambridge Syllabus Reference: 9701/31/M/J/22/Q2 · [20 Marks]

Chemicals and Apparatus Provided:

- FA 2 is $0.100 \text{ mol dm}^{-3}$ sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$.
- FA 3 is 1.00 mol dm^{-3} hydrochloric acid, HCl .
- Conical flask (100 cm^3) placed over a white tile marked with a black cross (X).
- Distilled water and stopwatch.

Reaction Equation:



Experimental Procedure:

1. In 5 separate experiments, measure the volumes of FA 2 and distilled water indicated in the table into the conical flask placed over the cross.
2. Add 5.0 cm^3 of 1.00 mol dm^{-3} HCl (FA 3), swirl once, and start the stopwatch.
3. Look vertically down through the flask from above.
4. Stop the watch the instant the black cross is completely obscured by the yellow precipitate of sulfur.
5. Record the reaction time, t , to the nearest second, and calculate $1/t$ (which is directly proportional to rate).

(a) Complete the experimental table below by recording time, t , and calculating $1/t$ (to 3 significant figures). [5]

Expt	Vol $\text{Na}_2\text{S}_2\text{O}_3 / \text{cm}^3$	Vol Water / cm^3	$[\text{Na}_2\text{S}_2\text{O}_3] / \text{mol dm}^{-3}$	Time, t / s	Rate ($1/t$) / s^{-1}
1	50.0	0.0	0.100		
2	40.0	10.0	0.080		
3	30.0	20.0	0.060		
4	20.0	30.0	0.040		
5	10.0	40.0	0.020		

(b) Plot a graph of Rate ($1/t$, vertical axis) against volume of $\text{Na}_2\text{S}_2\text{O}_3$ (horizontal axis). Draw the line of best fit. [5]

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(c) Using your graph, deduce the order of reaction with respect to sodium thiosulfate. Explain your deduction. [3]

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(d) Explain why the total volume in the conical flask was kept constant at 55.0 cm^3 in all 5 experiments. [2]

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(e) State one toxic gas produced in this reaction and the safety precaution required when disposing of the mixture. [3]

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(f) State the main human error in determining the endpoint in this investigation and suggest an instrument that would eliminate this subjectivity. [2]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Physical Chemistry Practical 4: Reaction Kinetics

Question 1: Continuous Monitoring of Reaction Rate: Magnesium and Hydrochloric Acid [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Table formatted correctly, all readings recorded to 0.5 cm ³ , smooth progressive increase leveling off to constant volume [4].	[4]
(b)	Axes labeled with quantity and unit (Volume of gas / cm ³ , Time / s) [1] Appropriate linear scale covering >= 50% of grid [1] All points plotted accurately within 0.5 small square [1] Smooth curve of best fit drawn without kinks or double lines [1] Tangent drawn accurately at t = 0 s [1] Initial rate calculated from gradient: Gradient = V / t with correct units (cm ³ s ⁻¹) [1]	[6]
(c)	Initial rate recorded with correct unit: ~2.00 cm ³ s ⁻¹ (allow 1.80 to 2.20 cm ³ s ⁻¹). [1]	[1]
(d)	As the reaction proceeds, reactant particles (Mg and H ⁺ ions) are consumed. The concentration of H ⁺ ions decreases, leading to fewer collisions per unit time between reactant particles, lowering collision frequency and reducing reaction rate. [2]	[2]
(e)	Max volume = 52.0 cm ³ = 0.0520 dm ³ . Moles of H ₂ = 0.0520 / 24.0 = 2.167 x 10 ⁻³ mol. [2]	[2]
(f)	Moles of Mg = Moles of H ₂ = 2.167 x 10 ⁻³ mol. Mass of Mg = 2.167 x 10 ⁻³ x 24.3 = 0.0527 g. [2]	[2]
(g)	Gas escapes into the air between adding the magnesium strip and pushing the rubber bung into the flask neck [1]. Modification: Suspend the magnesium strip on a thread inside the flask clamped by the bung, or place the magnesium in a small test-tube inside the flask and invert/tilt to start the reaction after sealing [2].	[3]

Question 2: Disappearing Cross Kinetics: Effect of Concentration on Sodium Thiosulfate Rate [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	All 5 experiments completed, times recorded in integer seconds, 1/t calculated correctly to 3 s.f., times increase as concentration decreases [5].	[5]
(b)	Axes labeled with quantity and unit (1/t / s ⁻¹ , Volume of Na ₂ S ₂ O ₃ / cm ³) [1] Linear scales using >= 50% of grid [1] Points plotted accurately within 0.5 square [1] Straight line of best fit drawn passing through the origin (0,0) [2]	[5]
(c)	The order of reaction is 1 (first order) [1]. The graph of rate (1/t) against concentration (volume) is a straight line passing through the origin (0,0), demonstrating that rate is directly proportional to concentration: Rate proportional to [Na ₂ S ₂ O ₃] ¹ [2].	[3]
(d)	To ensure the volume of sodium thiosulfate used is directly proportional to its concentration in the reaction mixture, and to keep the depth of liquid in the flask constant so that the same mass of sulfur precipitate is required to obscure the cross each time. [2]	[2]
(e)	Toxic gas: Sulfur dioxide (SO ₂) [1]. Safety precaution: Carry out in a well-ventilated laboratory/fume cupboard; immediately pour the reaction mixture into a sodium carbonate / sodium hydrogencarbonate stop bath to neutralize SO ₂ and prevent it escaping into the room [2].	[3]
(f)	Human judgment of the exact moment the cross disappears is subjective and varies between observers [1]. Instrumental replacement: A colorimeter or light sensor / datalogger connected to a computer to measure transmitted light intensity objectively [1].	[2]